
Final Exam

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MA 542 Section A1

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Problem 1

Let $X := \{\sigma(g) \mid \sigma \in G\}$. Since $|\text{Gal}(E/F)| = [E : \mathbb{Q}] = 6$, we immediately have that $|X| = 6$, provided that all such $\sigma(g)$ are distinct, and thus it suffices to show linear independence of $\{\sigma(g) \mid \sigma \in G\}$ (and that they are indeed distinct). Note that

$$0 = \zeta^3 - 1 = (\zeta - 1)(\zeta^2 + \zeta + 1).$$

Since $\zeta \neq 0$, we have that $\zeta^2 + \zeta + 1 = 0$. Using this explicit computation gives

$$\begin{aligned} 1(\theta) &= \theta \\ x(\theta) &= 1 + \zeta \sqrt[3]{2} + \zeta^2 \sqrt[3]{2}^2 + \zeta + \zeta^2 \sqrt[3]{2} + \zeta^3 \sqrt[3]{2}^2 \\ &= 1 + \zeta \sqrt[3]{2} - (1 + \zeta) \sqrt[3]{2}^2 + \zeta - (1 + \zeta) \sqrt[3]{2} + \sqrt[3]{2}^2 \\ &= 1 - \sqrt[3]{2} + \zeta - \zeta \sqrt[3]{2}^2 \\ x^2(\theta) &= 1 - \zeta \sqrt[3]{2} + \zeta - \zeta^3 \sqrt[3]{2}^2 \\ &= 1 - \sqrt[3]{2}^2 + \zeta - \zeta \sqrt[3]{2} \\ t(\theta) &= 1 + \sqrt[3]{2} + \sqrt[3]{2}^2 - (\zeta + 1) - (\zeta + 1) \sqrt[3]{2} - (\zeta + 1) \sqrt[3]{2}^2 \\ &= -\zeta - \zeta \sqrt[3]{2} - \zeta \sqrt[3]{2}^2 \\ tx(\theta) &= t \left(1 - \sqrt[3]{2} + \zeta - \zeta \sqrt[3]{2}^2 \right) \\ &= 1 - \sqrt[3]{2} - (\zeta + 1) + (\zeta + 1) \sqrt[3]{2}^2 \\ &= -\zeta - \sqrt[3]{2} + \sqrt[3]{2}^2 + \zeta \sqrt[3]{2}^2 \\ tx^2(\theta) &= t \left(1 + \zeta - \zeta \sqrt[3]{2} - \sqrt[3]{2}^2 \right) \\ &= 1 - (\zeta + 1) + (\zeta + 1) \sqrt[3]{2} - \sqrt[3]{2}^2 \\ &= -\zeta + \zeta \sqrt[3]{2} + \sqrt[3]{2} - \sqrt[3]{2}^2. \end{aligned}$$

We trivially have that

$$\mathcal{B} := \left\{ 1, \sqrt[3]{2}, \sqrt[3]{2}^2, \zeta, \zeta \sqrt[3]{2}, \zeta \sqrt[3]{2}^2 \right\}$$

spans the space, and since we are given the space is 6-dimensional, we immediately have that this set is linearly independent, and thus a basis. We can form the matrix

$$A := \begin{pmatrix} \theta & x(\theta) & x^2(\theta) & t(\theta) & tx(\theta) & tx^2(\theta) \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & -1 & 1 \\ 1 & 0 & -1 & 0 & 1 & -1 \\ 1 & 1 & 1 & -1 & -1 & -1 \\ 1 & 0 & -1 & -1 & 0 & 1 \\ 1 & -1 & 0 & -1 & 1 & 0 \end{pmatrix}$$

in the basis $\mathcal{B}$ (with ordering of elements as in our description of the set), and we know from linear algebra that since $\mathcal{B}$ is a basis, we will have that X is a basis if and only if A is $\mathbb{Q}$ -invertible. Since $\mathbb{Q}$ is a field, it suffices to compute the determinant of A .

```

1 #!/bin/python3
2
3 import numpy as np
4
5 a = np.array([[1,1,1,0,0,0],
6               [1,-1,0,0,-1,1],
7               [1,0,-1,0,1,-1],
8               [1,1,1,-1,-1,-1],
9               [1,0,-1,-1,0,1],
10              [1,-1,0,-1,1,0]])
11 det = np.linalg.det(a)
12 print(det)

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1 ~/Documents/code_projects > ./deter.py
2 >>> 27.0

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Since $27 \neq 0$, A is invertible, and X is a basis for $E/\mathbb{Q}$.

Problem 2

1. By the Galois correspondence, there exists a subgroup $H \subseteq \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$ which is the group that fixes $\mathbb{Q}(\zeta_n + \bar{\zeta}_n)$. One easily shows that the fixed field of $\langle c \rangle$, for c the element corresponding to complex conjugation, is $\mathbb{Q}$ adjoined with any element fixed by c . Since complex conjugation now fixes both $\mathbb{Q}$, $\zeta_n \bar{\zeta}_n$, and $\zeta_n + \bar{\zeta}_n$, all of which live in $\mathbb{Q}(\zeta_n + \bar{\zeta}_n)$, we have $H = \langle c \rangle$. Since $\bar{\bar{z}}_n = z$ for all $z \in \mathbb{C}$, we know $|c| = 2$, and thus $|\langle c \rangle| = 2$. Finally, we know that

$$[\mathbb{Q}(\zeta_n) : \mathbb{Q}(\zeta_n + \bar{\zeta}_n)] = |H| = 2.$$

2. We are given that $\mathbb{Q}(\zeta_n)$ extends $\mathbb{Q}(\zeta_n + \bar{\zeta}_n)$; in particular, $\mathbb{Q}(\zeta_n + \bar{\zeta}_n) \subseteq \mathbb{Q}(\zeta_n)$. We also have $\mathbb{Q} \subseteq \mathbb{R}$, and by our understanding of complex conjugates, $\zeta_n + \bar{\zeta}_n \in \mathbb{R}$. Therefore, $\mathbb{Q}(\zeta_n + \bar{\zeta}_n) \subseteq \mathbb{R}$, and we have $\mathbb{Q}(\zeta_n + \bar{\zeta}_n) \subseteq \mathbb{Q}(\zeta_n) \cap \mathbb{R}$. It suffices to prove the converse.

For any $x \in \mathbb{Q}(\zeta_n) \cap \mathbb{R}$, we have that the conjugate $\bar{x} = x$, since $x \in \mathbb{R}$. It follows that x is an element of the fixed field of the complex conjugate, which we found to be $\mathbb{Q}(\zeta_n + \bar{\zeta}_n)$ in the previous problem. Therefore, $\mathbb{Q}(\zeta_n) \cap \mathbb{R} \subseteq \mathbb{Q}(\zeta_n + \bar{\zeta}_n)$.

3. The minimal polynomial of $\mathbb{Q}(\zeta_n)$ is the n th cyclotomic polynomial, which takes as its roots all n th primitive roots of unity, which are of the form

$$e^{i2\pi k/n}, \quad k \in \mathbb{Z} \setminus n\mathbb{Z}.$$

It follows that for all $k \notin n\mathbb{Z}$, we have $-k \notin n\mathbb{Z}$, and thus

$$\bar{\zeta}_n^k = \overline{e^{i2\pi k/n}} = e^{-i2\pi k/n}.$$

Therefore, for any primitive n th root of unity ζ_n^k (and thus a root of the minimal polynomial), the complex conjugate is still a primitive n th root of unity, and thus also a root of the minimal polynomial. Therefore, complex conjugation permutes the roots of the minimal polynomial, and is thus an element of the Galois group.

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4. Note that for all $z \in \mathbb{C}$, we know $z\bar{z} = |z|^2$. Also note that $|e^{i\theta}| = 1$ for all $\theta \in \mathbb{R}$. Since ζ_n takes this form, we have that

$$\zeta_n \bar{\zeta}_n = |\zeta_n|^2 = 1^2 = 1.$$

Therefore, $\bar{\zeta}_n = \zeta_n^{-1}$. We know that $\sigma_k \in \text{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$ if and only if $\sigma(\zeta_n) = \zeta_n^k$ for some k . We can then simply identify σ_k with $[k] \in U(n)$, and we find that since $\bar{\zeta}_n = \zeta_n^{-1}$, we must have $[-1] \in U(n) = n - 1 \pmod{n}$ is the element corresponding to complex conjugation.

Problem 3

We prove that there exists a normal subgroup which is solvable, whose quotient is also solvable. This suffices to prove that the group is solvable by an in-class result.

Let $|G| = p_1 p_2 p_3$, and let n_i denote the number of Sylow p_i -subgroups. First, we want to show there exists a normal subgroup of G . Suppose for contradiction that $n_1, n_2, n_3 > 1$. By Sylow's theorem, $n_3 | p_1 p_2$, and thus $n_3 \in \{1, p_1, p_2, p_1 p_2\}$. By assumption, $n_3 \neq 1$. By Sylow's theorem, $n_3 \equiv 1 \pmod{p_3}$, and since $p_1, p_2 < p_3$ by assumption, we have

$$p_1 \equiv 1 \pmod{p_3}, \quad p_2 \equiv 1 \pmod{p_3}.$$

Since p_1, p_2 are prime, $p_1, p_2 \neq 1$, and thus $n_3 \neq p_1, p_2$. We thus have that $n_3 = p_1 p_2$ is forced.

Similarly, we have by Sylow's theorem that $n_2 | p_1 p_3$, and thus $n_2 \in \{1, p_1, p_3, p_1 p_3\}$. Since $p_1 < p_2$, for the same reason we have $n_2 \neq p_1$. By assumption, $n_2 \neq 1$. Thus, $n_2 \in \{p_3, p_1 p_3\}$ is forced, and thus $n_2 \geq p_3$. By the same logic, we find that $n_1 \in \{p_2, p_3, p_2 p_3\}$, and thus $n_1 \geq p_2$.

Note that all non-identity elements of the Sylow p_3 -subgroups have order p_3 , and similarly for Sylow p_1 - and p_2 -subgroups. Since $p_1 \neq p_2 \neq p_3$, it follows that that these elements must all be distinct from each other. Moreover, each Sylow p_i -subgroup must be disjoint from any other Sylow p_i -subgroup except for the identity, since every non-identity element is a generator.

Since there are $p_1 p_2$ Sylow p_3 -subgroups, and since each one has at least $p_3 - 1$ unique elements, we have that there are at least $(p_1 p_2)(p_3 - 1)$ unique (aka non-identity) elements given by Sylow p_3 -subgroups. Similarly, we have at least $p_2(p_1 - 1)$ unique elements in Sylow p_1 -subgroups, and at least $p_3(p_2 - 1)$ Sylow p_2 -subgroups. We thus have

$$p_1 p_2 p_3 - n_3(p_3 - 1) - n_2(p_2 - 1) - n_1(p_1 - 1) - 1 \geq p_1 p_2 p_3 - (p_1 p_2)(p_3 - 1) - p_3(p_2 - 1) - p_2(p_1 - 1) - 1 \geq 0,$$

where the -1 represents the identity. Distributing we have

$$\begin{aligned} &= p_1 p_2 p_3 - p_1 p_2 p_3 + p_1 p_2 - p_3 p_2 + p_3 - p_2 p_2 + p_2 - 1 \\ &= p_3 + p_2 - p_3 p_2 - 1 \\ &= (p_3 - 1)(1 - p_2). \end{aligned}$$

Since $p_3, p_2 > 1$, we have $p_3 - 1 > 0$ and $1 - p_2 < 0$, and thus $(p_3 - 1)(1 - p_2) < 0$, meaning the group has a negative amount of elements a contradiction. Therefore, there exists some i such that $n_i = 1$. By Sylow's theorem, this group must be normal.

Let $H \triangleleft G$ be a normal Sylow p_i -subgroup. Since H has order p_i for some i , it has prime order, and is thus cyclic, and is thus abelian. Therefore, it is solvable. It thus suffices to show that G/H is solvable. Note that G/H has order

$$\left| \frac{G}{H} \right| = \frac{|G|}{|H|} = \frac{p_1 p_2 p_3}{p_i} = p_a p_b$$

for some $a, b \neq i$, and thus it suffices to show groups with order $|A| = p_a p_b$ are solvable for any primes p_a, p_b .

Let A be a group of order $p_a p_b$. By Lagrange's theorem, any proper nontrivial subgroup has order p_a or p_b , and thus any proper nontrivial subgroup has prime order, and is thus cyclic, and is thus abelian, and is thus solvable. Moreover, by Cauchy's theorem, for any prime divisor p of $|A|$, there exists an element of order p , and thus a cyclic subgroup of order p . We thus have that there exist subgroups of order p_a and p_b in A .

Let $N \subseteq A$ be one such subgroup, and without loss of generality let $p_b > p_a$, and fix $|N| = p_b$. By Sylow's theorem, the number of Sylow p_b -subgroups n_b satisfies $n_b \equiv 1 \pmod{p_b}$ and $n_b | p_a$. The first criterion forces $n_b = np_b + 1$, and since $p_b > p_a$, the second criterion forces $n = 0$, and $n_b = 1$. Therefore, N is normal in A .

We can thus quotient by N , and obtain a group of order

$$\left| \frac{A}{N} \right| = \frac{|A|}{|N|} = \frac{p_a p_b}{p_b} = p_a.$$

Since this is a group of prime order, it is cyclic, and thus abelian, and thus solvable. Therefore, A is solvable, concluding the proof.

In general, G need not be abelian. Dihedral groups are solvable, but D_{15} has order $30 = 2 \cdot 3 \cdot 5$, and is not abelian.

Problem 4

4.1

Let $\alpha \in E$. Note that the fixed field F^G by G is simply F , and we have $x \in F$ if and only if x is fixed by all elements of G . Let $\sigma \in G$, and since σ is a ring homomorphism, composition distributes over addition, and we have

$$\sigma(T(\alpha)) = \sigma\left(\sum_{g \in G} g(\alpha)\right) = \sum_{g \in G} (\sigma \circ g)(\alpha).$$

Note that $\sigma \circ g$ is left-multiplication by an element of G , and by Cayley's theorem, every group acts transitively on itself by left-multiplication. Equivalently, left multiplication by σ is a bijection on G , and thus composing on the left by σ simply amounts to a permutation of the order of the summation. We thus have

$$\sum_{g \in G} (\sigma \circ g)(\alpha) = \sum_{g \in G} g(\alpha).$$

Therefore, σ fixes $T(\alpha)$ for all $\sigma \in G$, we have that $T(\alpha) \in F$. Note that since σ also distributes over multiplication, since it is a ring homomorphism, the exact same proof suffices for $N(\alpha)$ since multiplication is commutative in fields, and the permutation of the ordering is once again immaterial.

For surjectivity, let $\alpha \in F$. Note that since F is a field, it is closed under addition, multiplication, and inverses, and since $\text{char } F$ does not divide $|G|$, we have that $\sum_{|G|} 1 \neq 0$, and thus

$$\beta := \left(\sum_{|G|} 1\right)^{-1} \alpha \in F.$$

Since $\beta \in F$, it is fixed by all $g \in G$, and thus

$$T(\beta) = \sum_{g \in G} g(\beta) = \sum_{g \in G} \beta = \sum_{|G|} \left(\left(\sum_{|G|} 1 \right)^{-1} \alpha \right) = \left(\sum_{|G|} 1 \right)^{-1} \left(\sum_{|G|} 1 \right) \alpha = \alpha.$$

Therefore, there exists an element $\beta \in F \subseteq E$ such that $T(\beta) = \alpha$, and T is surjective.

4.2

Let $E = F(\alpha)$, and let $q(x) \in F[x]$ be the polynomial

$$q(x) = \sum_{i=0}^n a_i x^i, \quad a_n = 1.$$

Suppose q splits in E . We can then write

$$q(x) = \prod_{i=1}^n (x - r_i)$$

for $\{r_i\}_{i=1}^n$ the collection of roots of q . Since E is Galois, the Galois group acts transitively on the roots of q . In particular, given the fixed root α , for all r_i there exists $g_i \in \text{Gal}(E/F)$ such that $r_i = g_i(\alpha)$. Moreover, since q is the minimal polynomial of E , we have $|\text{Gal}(E/F)| = \deg q = n$. It follows that

$$q(x) = \prod_{i=1}^n (x - g_i(\alpha)).$$

It is clear that the constant term a_0 is thus

$$a_0 = \prod_{i=1}^n -g_i(\alpha) = (-1)^n \prod_{g \in G} g(\alpha) = (-1)^n N(\alpha).$$

Since $(-1)^n$ is either -1 or 1 , it is its own inverse, and thus moving it over to the other side gives $N(\alpha) = (-1)^n a_0$.

Similarly, the x^{n-1} term easily seen to be given by the sum over the product of x^{n-1} with each $g(\alpha)$:

$$a_{n-1} x^{n-1} = \sum_{g \in G} -g(\alpha) x^{n-1} \implies a_{n-1} = - \sum_{g \in G} g(\alpha) = -T(\alpha).$$

Thus, $T(\alpha) = -a_{n-1}$.

4.3

Since g is a field automorphism for all $g \in G$, we have

$$T(x) = \sum_{g \in G} g(y - \sigma(y)) = \sum_{g \in G} g(y) - g(\sigma(y)) = \sum_{g \in G} g(y) - \sum_i \dots$$

By Cayley's theorem, every group acts transitively on itself by right multiplication, and thus composing on the right with σ is once again a bijection, and amounts to a reordering of the summation terms. We thus have that the above summation simply equals

$$\sum_{g \in G} g(y) - \sum_{g \in G} g(y) = 0.$$

Therefore, $T(x) = 0$.

4.4

Since g is a ring automorphism for all $g \in G$, it preserves multiplication as well. By Cayley's theorem, right composition by σ is a bijection. Since multiplication is commutative, inversion distributes over multiplication. Using all of these facts, we obtain

$$\begin{aligned} N(x) &= \prod_{g \in G} g(y(\sigma(y))^{-1}) = \prod_{g \in G} g(y)g(\sigma(y))^{-1} = \left(\prod_{g \in G} g(y) \right) \left(\prod_{g \in G} (g \circ \sigma)(y)^{-1} \right) \\ &= \left(\prod_{g \in G} g(y) \right) \left(\prod_{g \in G} g(y)^{-1} \right) = \left(\prod_{g \in G} g(y) \right) \left(\prod_{g \in G} g(y) \right)^{-1} = 1. \end{aligned}$$

Therefore, $N(x) = 1$.

4.5

Since $\mathbb{Q}(i)$ is 2-dimensional as a vector space over $\mathbb{Q}$, the Galois group must have order 2. Moreover, note that complex conjugation is trivially an element of the Galois group, and thus σ is complex conjugation.

We have

$$\begin{aligned} x &= \frac{y}{\bar{y}} \\ &= \frac{c + di}{c - di} \\ &= \frac{c + di}{c - di} \cdot \frac{c + di}{c + di} \\ &= \frac{(c + di)(c + di)}{(c + di)(c - di)} \\ &= \frac{c^2 + 2cdi - d^2}{c^2 + d^2} \\ &= \frac{c^2 - d^2}{c^2 + d^2} + \frac{2cd}{c^2 + d^2}i. \end{aligned}$$

We thus have

$$a = \operatorname{Re} x = \frac{c^2 - d^2}{c^2 + d^2}, \quad b = \operatorname{Im} x = \frac{2cd}{c^2 + d^2}.$$

4.6

By our previous result, we have

$$x = \frac{A + Bi}{C}.$$

Applying the fact that $N(x) = 1$, we have

$$\begin{aligned} 1 &= x \cdot \bar{x} \\ &= \frac{A + Bi}{C} \cdot \frac{A - Bi}{C} \\ &= \frac{A^2 + B^2}{C^2} \\ \implies C^2 &= A^2 + B^2. \end{aligned}$$